Supplementary Information: Identifiability of independent sampling units in publicly deposited metabolomics data

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Abstract

Supplementary Methods S1–S5 and Supplementary Tables S1–S3 referred to in the main text. All file names, hashes and commit identifiers refer to the public repository cited in the main text's Data Availability statement.

Supplementary Methods S1: MetaboLights census, universe, access boundary and rules

Universe.

A separately locked enumeration audit compared the MetaboLights REST study list with the public FTP mirror on 25 October 2026. The REST list (3,584 accessions) was a subset of the mirror (3,596 folders); the 12 mirror-only accessions were recorded as unresolved and not screened. Two accessions used as development exemplars were excluded, leaving a screening universe of 3,582 accessions.

Access boundary.

For each accession the census requested the directory index and every investigation, sample and assay file it listed (`i_`, `s_`, `a_`). Metabolite assignment files, data folders, raw and derived data and redirect destinations were refused by an allow-list guard. Requests were journalled with their HTTP status, size and declared length.

Rules.

Studies were screened in a fixed order: organism (M1: human samples only), sample-sheet integrity and quality-control rows (M2: rows classified as blanks, pooled or reference material, or quality control were removed; a study with no biological rows left was unresolved), duplicate sample names (M3), a single sample matrix (M4), and the independence test (M5). An independent subject was defined as a human individual contributing exactly one biological sample to the chosen assay, and M5 had two affirmative routes. R1 required a person-identifier column on the source node (for example `Characteristics[Subject ID]`, `[Participant]`, `[Donor]` or `[Patient]`, by a pattern frozen in code), present on every biological row, with exactly one sample per identifier. R2 required two layers. Layer 1 classified the design from the study design descriptors and description: at least one cross-sectional or case-control term and no longitudinal, repeated-measures, crossover, follow-up or pre/post term. Layer 2 required a sentence in the study description matching one of four frozen templates and containing no negation or plural-sampling token. Templates T1 and T2 state one sample per participant explicitly ("one sample was collected from each participant"). Templates T3 and T4 state a number of distinct participants, accepted only if the integer equalled the number of biological sources exactly. Both routes also required one sample per source and at least 50 sources. The strict criterion (L1) counted R1 and T1/T2 evidence; a sensitivity count added T3/T4 evidence, because equal cardinalities do not prove a one-to-one mapping. Studies that failed M5 but had at least 50 distinct source labels with one sample each ("source-unique") or at least 50 samples with a degenerate source column were recorded as unresolved candidates for documentary assessment.

Execution.

The census ran in three locked invocations: a full run (code v0.1.6) and two disclosed retry passes for transport failures and a rescreen of studies whose sample rows had all been classified as quality control (v0.1.7 and v0.1.8). The final record for each accession is the latest one.

Supplementary Methods S2: Metabolomics Workbench secondary audit, provenance of the 104 studies

Table S1 reproduces the selection steps, as recorded in the closure record of the earlier screen, that produced the 104 studies whose `factors` endpoint was audited. The counts are those of that record and were not recomputed here.

Table S1 Provenance of the 104 Metabolomics Workbench studies (from the closure record of the earlier screen; counts as recorded there)

Step	n
Public studies in the Workbench species and source lists, 17 October 2026	4,635
of which human	1,906
E1: not human, or no species record (excluded)	2,716 / 13
E2: studies previously opened by the authors' earlier work (excluded)	216
PRE / PRE-N: fewer than 60 reported metabolites / fewer than 50 samples (excluded)	966 / 501
Pass PRE and PRE-N	223
E4: analysis attribution unresolved (excluded)	20
E5: repeated measures / multiple matrices / $n < 50$ after attribution (excluded)	48 / 41 / 10
Retained for this audit: eligible 72, E6 (metabolite-class rule, not applied here) 28, ambiguous naming 4	**104**

Supplementary Methods S3: Documentary assessment, sources, reading views, document freeze and grading rules

Sources.

Four sources were admissible: the deposit's investigation file (S1); publications named in the deposit by PubMed identifier or DOI, retrieved as full text from Europe PMC (S2); the first five publications, by first publication date, returned by a frozen Europe PMC full-text search for the exact accession string, usable only if their text contained the accession (S3); and author-supplied copies of S2 publications that were not open access, registered and hashed before the document freeze. No author was contacted.

Value-blind reading views.

Extractors read only a deterministic reading view of each publication: the title; the abstract with any Results, Findings, Conclusions or Discussion parts removed; methods-type sections; and participant-characteristics tables. Results and Discussion sections, figures and supplementary files were excluded. For PDF copies a heuristic view (front matter and the methods block) was used, and records citing it carry a flag.

Document freeze.

After retrieval, every document and view was listed with its hash in an evidence inventory, which the author signed before any evidence was read. No document could be added after the signature.

Table S2 Grading rules for the documentary assessment, applied in order (protocol v0.2, §9)

Order	Condition	Grade
0	failed quotation check or schema error	invalid (returned)
1	assay membership (Q2) not established	UNRESOLVED
2	Q4 states repeated sampling, or Q7 shows repeated samples	CONTRADICTED
3	Q1: samples not from human participants	CONTRADICTED
4	Q1 not stated	UNRESOLVED
5	no participant count scoped to the analysed set	UNRESOLVED
6–7	stated count ≠ deposited biological rows	UNRESOLVED
8	count or assay link supported only by the deposit itself	UNRESOLVED
8b	assay link only from S3 papers without the accession in the quote	UNRESOLVED
9	one sample per participant stated in a publication (Q4 or Q7)	STRICT
10	otherwise	COUNT (sensitivity)

Supplementary Methods S4: Integrity, access and data handling

The census, the Workbench audit and the documentary assessment were each executed under a signed lock that fixed the code, protocol and inputs by hash; code refused to run against any other version. Deviations and implementation limitations are recorded in the result records and summarised in the Results. Participant-level information was not recorded beyond counts. Pseudonymous source labels collected by the census are held in a checkpoint file that is not published. Full texts of publications are not redistributed.

Supplementary Methods S5: Author review protocol and deviations

Protocol. `T1_HUMAN_REVIEW_PROTOCOL_v1.0.md` (SHA-256 799865e8…f509) was hashed on 31 October 2026 before the selection was computed. Component A: sampling frame = the 17 tranche studies with at least one publication reading view that the author had not adjudicated; selection = the first 10 by SHA-256 of `docrec-T1-human-review-v1|<accession>`; questions Q3 (scope, count, quotation) and Q4 (category, quotation) with the frozen categories; agreement rules: exact category match for Q3 scope and Q4, integer equality for Q3 count where both chose `selected_assay`. Component B: all 12 final CONTRADICTED records; verdict confirmed / disputed / unresolved with a verbatim quotation.

Deviations. (1) The intended independent reviewer could not be recruited; the author performed both components. (2) Before Component A the author had seen a second AI model's answers for the same ten studies, produced in an earlier attempt that was withdrawn and relabelled as an AI extraction (`T1_SECOND_MODEL_A_sheet.csv`, provenance note in the repository); the author had not seen the original extractor's answers or quotations. (3) The author's sheets were

assembled by script after review; their timestamps are assembly artefacts. (4) Component B was performed by the author who had adjudicated the same records on 31 October 2026. The author's attestations are recorded verbatim in the result notes `T1_HUMAN_REVIEW_A_RESULT_v1.0.md` and `T1_HUMAN_REVIEW_B_RESULT_v1.0.md`.

Supplementary Note: implementation limitations of the frozen code

The frozen implementation had three limitations, each disclosed in the result records. One reading view included a differential-metabolite table whose caption used an abbreviation that escaped the exclusion pattern; nothing was cited from it. The PDF view heuristic did not recognise "STAR+METHODS" headings; no graded record depended on such a block. The count check accepted an integer from a deposit quotation when the participant-count question also cited a publication; in the one affected record the publication stated the disjoint group sizes that sum to the count.

Supplementary Table S3: Fields supporting V2 in the 19 quality-control-excluded deposits

Table S3 lists, for each of the 19 deposits, whether the V2 check passed and which deposited fields supported it.

Table S3 Deposits whose every sample row is labelled `experimental blank`, and the other deposited fields that identify the rows as study samples rather than blanks (V2). For MTBLS15165 the fields establish study material, not identifiable donors (V1 not established). Quotations are in the public evidence records.

Accession	V2	Fields supporting V2
MTBLS20	pass	investigation text (urine processing protocol); sample-file organism column
MTBLS148	pass	investigation text (urine collection from donors); sample-file factor values
MTBLS6260	pass	investigation text (plasma collection); sample-file time factors
MTBLS9021	pass	investigation text (baseline blood from patients); sample-file factor values
MTBLS11048	not established	none (cell-line rows)
MTBLS12684	pass	named publication; sample-file disease characteristic; investigation text
MTBLS12824	pass	investigation text; sample-file age-group and exercise factors
MTBLS13136	pass	investigation text; sample-file disease factor
MTBLS13319	not established	none (pooled plasma, method study)
MTBLS13382	pass	investigation text (DBS and serum collection); sample-file sample-type factor
MTBLS13634	pass	investigation text; sample-file treatment factor and group characteristic
MTBLS13827	pass	investigation text; sample-file treatment factor and group characteristic
MTBLS14358	pass	investigation text; sample-file organism column and group factor
MTBLS14376	pass	investigation text; sample-file group factor
MTBLS14421	pass	investigation text (serum from patients and volunteers); sample-file factor
MTBLS14426	pass	investigation text (urine and fasting serum collection); sample-file factor
MTBLS14865	pass	investigation text (urine before surgery or at examination); sample-file treatment factor
MTBLS15004	pass	investigation text (faecal samples from patients in two groups)
MTBLS15165	pass[†]	investigation text (stored platelet units); sample-file replicate factor

[†]Study material established; donors not described.